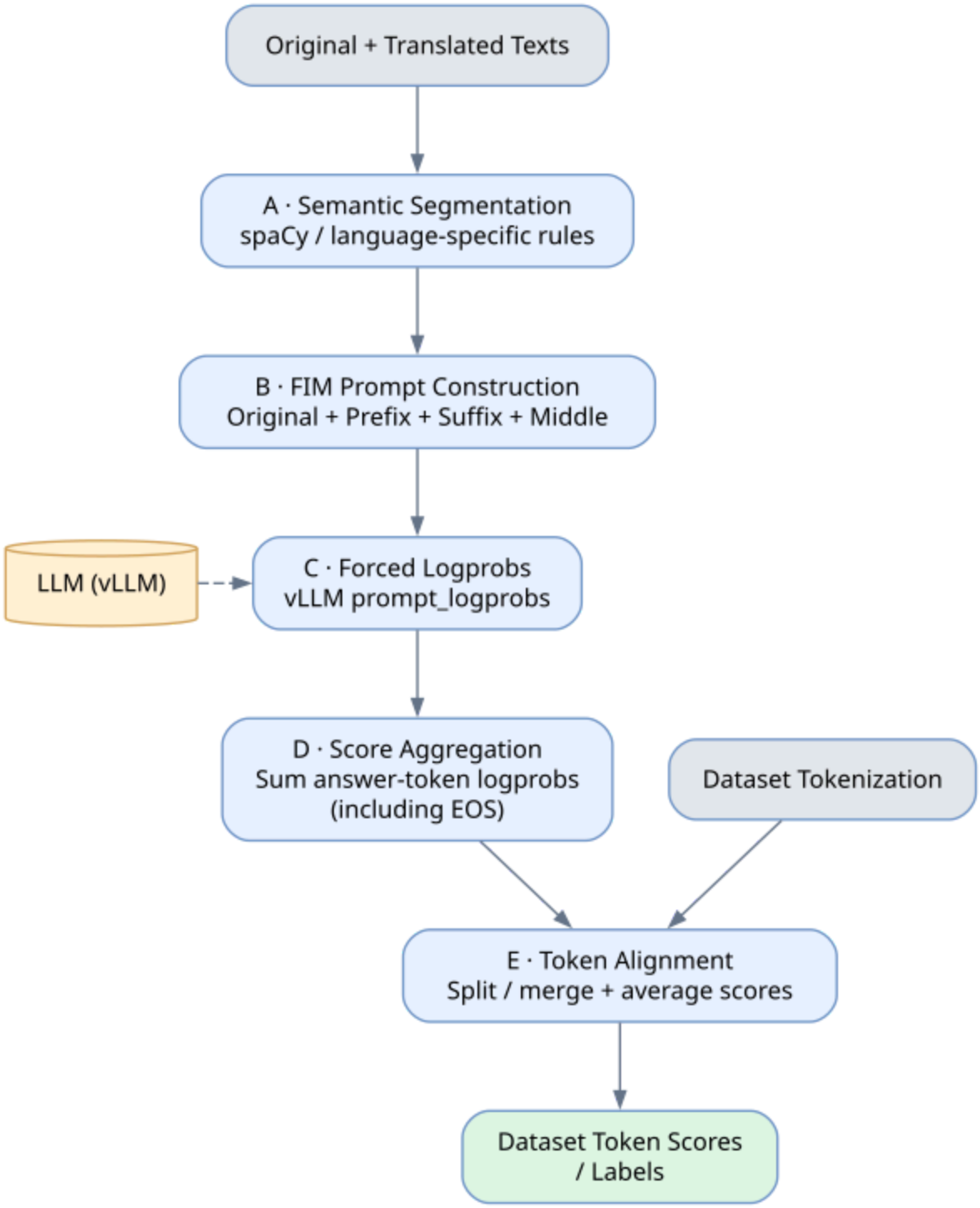
Short method description

YT-APERTUS-RSD

We address the RSD task of predicting per-token meaning divergence between parallel documents by treating it as a constrained fill-in-the-middle (FIM) problem: each document is first segmented into linguistically motivated semantic units (via language-specific spaCy pipelines that preserve multi-word expressions, named entities, negations, modals, etc.), then short spans are masked and the Apertus 70B model is forced to reconstruct them given both the original text and the surrounding translation context; the summed log-probabilities of the forced assistant tokens serve as a direct measure of translation fidelity, which are subsequently realigned to the official dataset tokenization by whitespace-free span matching and averaging.

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Original + Translated Texts
A · Semantic Segmentation
spaCy / language-specific rules
B · FIM Prompt Construction
Original + Prefix + Suffix + Middle
LLM (vLLM)
C · Forced Logprobs
vLLM prompt_logprobs
D · Score Aggregation
Sum answer-token logprobs
(including EOS)
Dataset Tokenization
E · Token Alignment
Split / merge + average scores
Dataset Token Scores
/ Labels

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